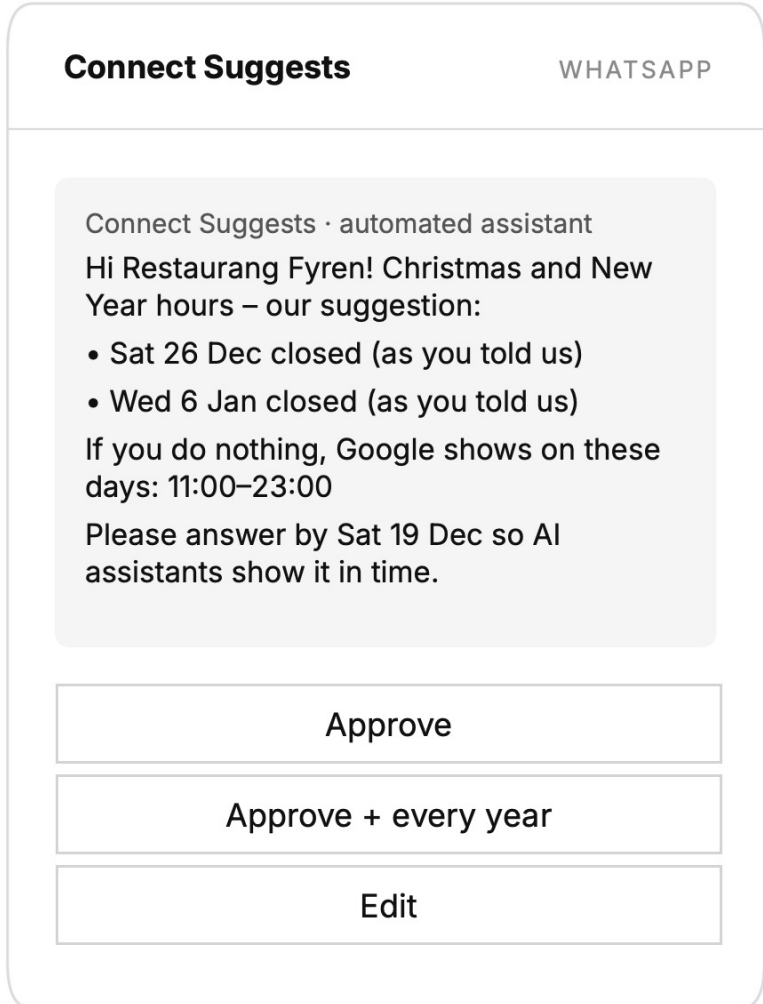
TEAM CONNECT · SINCE AI HACKATHON 2026

Fonecta's challenge “Connect Suggests – Customer Approves”

Connect Suggests

Connect suggests. The owner approves once. Every year it needs them less.

Right holiday opening hours on Google, Apple Maps and AI assistants, approved by the owner in their own chat, with no login.



Wrong holiday hours send customers to a locked door

Fonecta Connect publishes each business's hours to 35+ apps. Owners forget one-off holiday hours, so the apps keep showing normal hours.

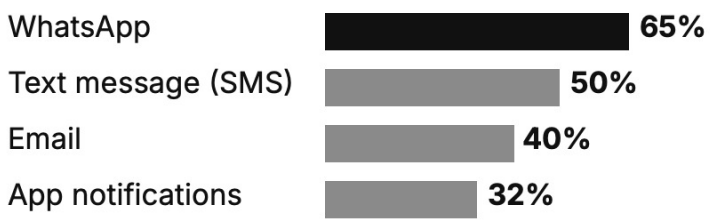
63%

of the changes that are needed are “shown open, actually closed”: the app says open, the door is locked.

53%

of suggestions in Fonecta's 2025 pilot got no reply (52% of owners never answered).

Reply rate per suggestion, by channel

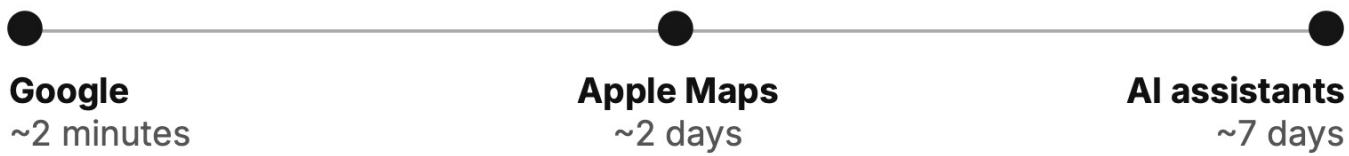


“Copy last year” can only see 44% of the needed changes



So Connect also reads two seasons back and the owner's other holidays.

Apps update at different speeds



So every message has an “answer by” date that gives the slowest app time.

All five steps of the brief, working end to end

Running on Fonecta's own stack: Yext, the service Fonecta Connect already publishes listings through, plus the owner's own chat.

1. **Detect**

   Rebuild what each app will show on every special day, and find the days that will be wrong.

2. **Suggest**

   Predict the real hours from the business's own past, with a reason and a confidence.

3. **Contact**

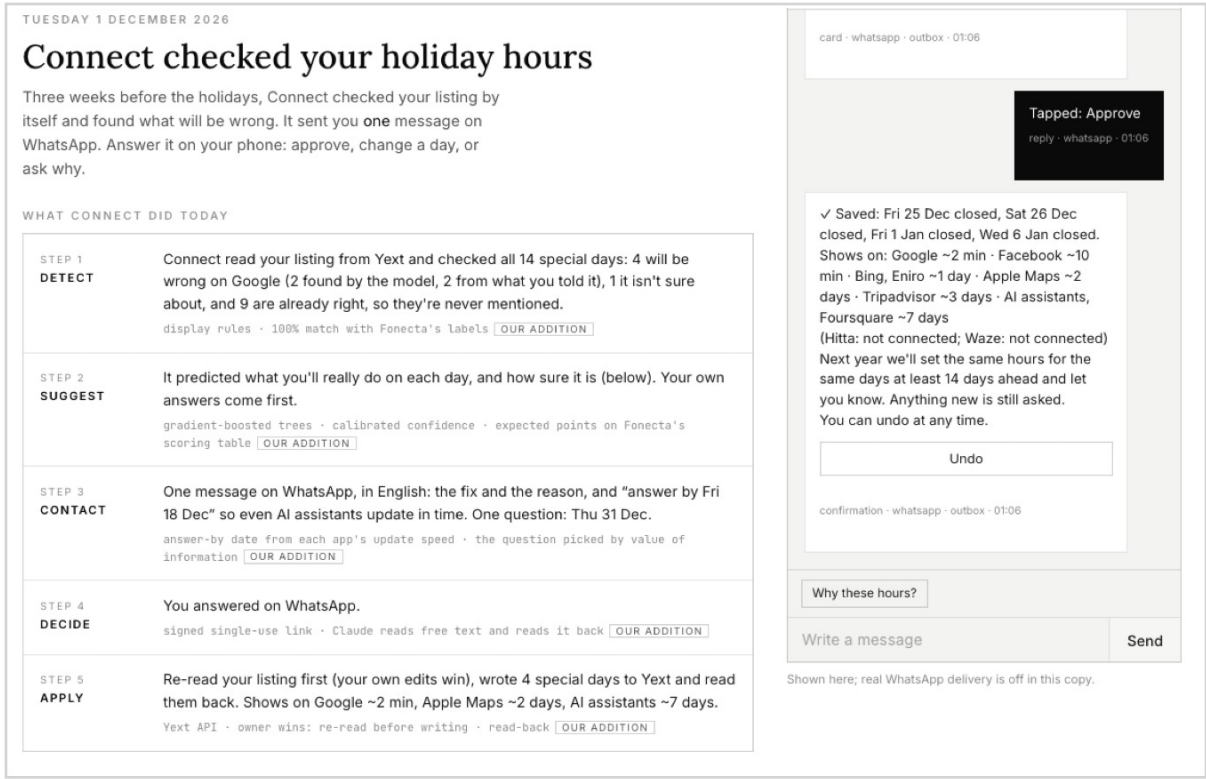
   One message per season on the owner's channel (WhatsApp, email or text), in their language.

4. **Decide**

   Approve, change a day or decline: one tap, or a reply in their own words. No login.

5. **Apply**

   Write through Yext, read it back, and say where it shows: Google ~2 min, Apple Maps ~2 days.



The demo app after the owner tapped Approve on WhatsApp: Connect wrote the hours, and the listing on Google is updated. A full run takes about 28 seconds.

Approve once, checked every year

For the owners who never reply. The brief asked for standing permission; we added a safety check before any answer is reused.

1. **One card, two equal buttons**

   Approve, or Approve + every year. One sentence explains the scope: the same days with the same hours; anything new is still asked.

2. **The model-agreement check**

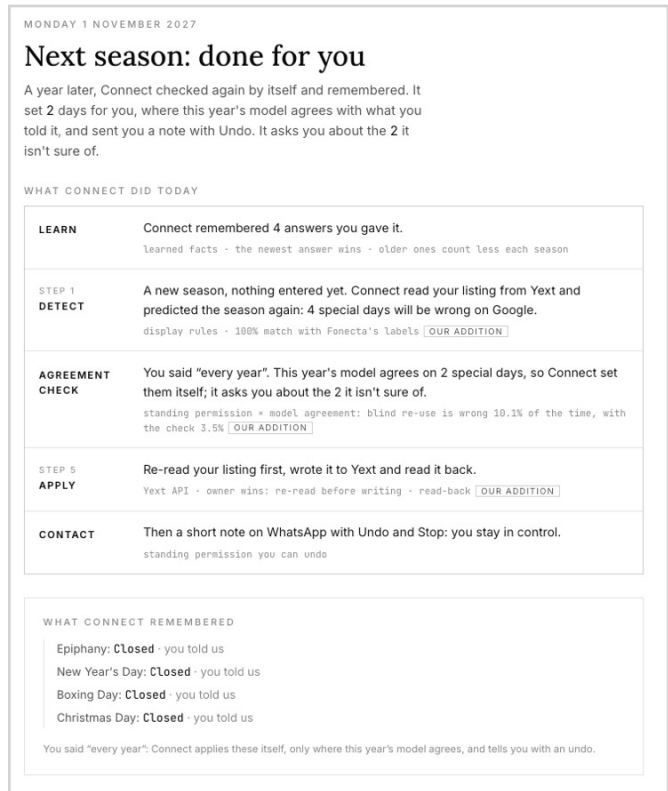
   Next season, Connect re-applies the owner's earlier answer only if this year's model gives it at least 50%. Otherwise it asks.

3. **A note with Undo**

   The owner gets a short note in their chat, with Undo and Stop. They stay in control.

10.1% → 3.5%

Re-using last year's answer blindly gets open and closed wrong 10.1% of the time. With the agreement check: 3.5%.



The demo, one year later: Connect set 2 days itself and asks about the 2 it isn't sure of.

The check makes automatic changes safe

Re-playing next season on businesses whose real hours are known. “Wrong” means open and closed mixed up.

10.1%

of automatic changes are wrong when last year's answer is re-used blindly

3.5%

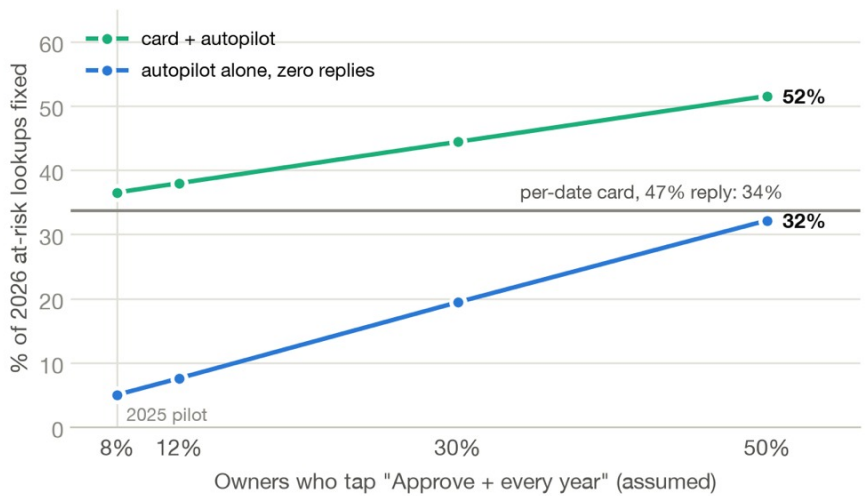
are wrong with the model-agreement check

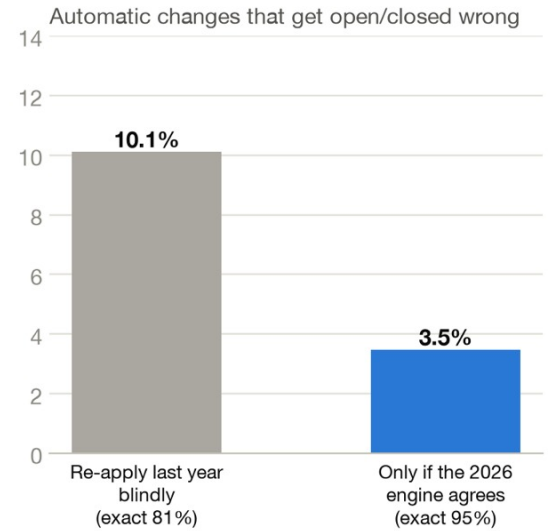
38.0% vs 33.8%

of next season's at-risk lookups fixed: with Autopilot vs the season card alone, if 12% of owners say “every year”

Season Autopilot reaches owners who never reply

Dev businesses; 2025 card approved, permission sampled at each rate (50 seeds). Grant rates are assumptions.





An assumption, not a measurement: the share of owners who say “every year” (12% is the rate in Fonecta's owner simulator). Only a live test can measure it. At-risk lookups = searches made on days the listing would be wrong.

61.67 points against the baseline's 54.49

Fonecta's own scoring script, with models that never saw the businesses they scored. Baseline = Fonecta's own example solution.

61.67
score, against 54.49 (95% range 60.4–62.9)

86%
of our suggestions are exactly right

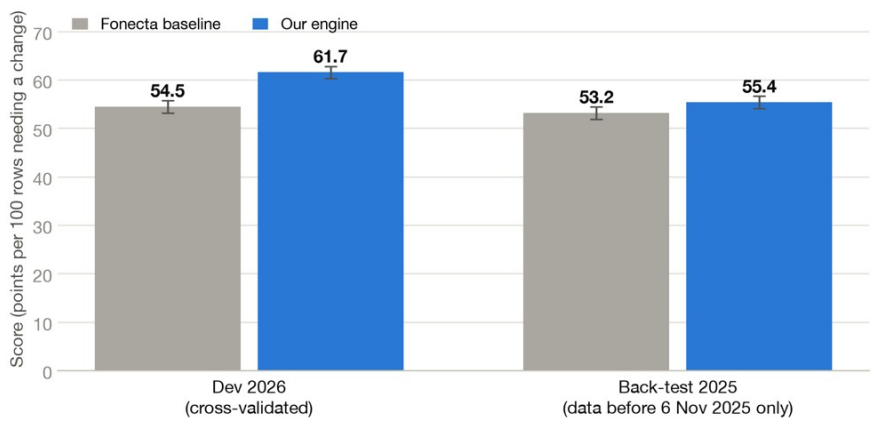
67%
of wrong days fixed, against 59%

219
harmful suggestions, against 224: fewer, but within noise

55.42
score in the 2025 back-test, against 53.23

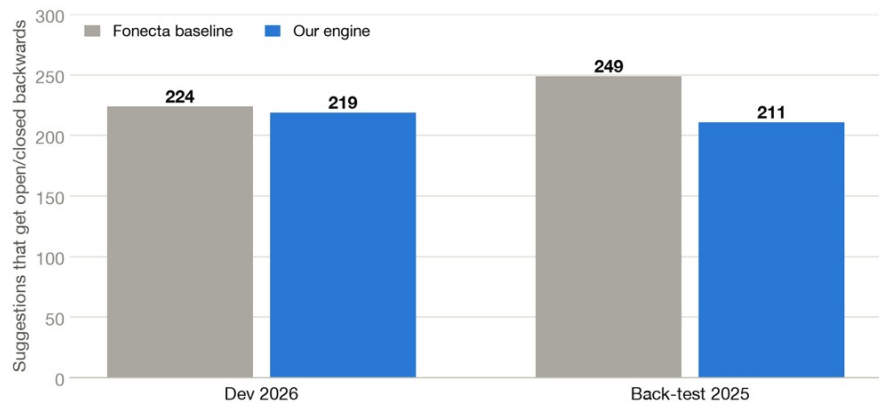
+7.2 points over Fonecta's baseline

Official scorer, dev businesses. Error bars: 95% interval at the size of the hidden test set.



More right answers, without more closed-door mistakes

Count of harmful suggestions on dev rows: "closed" for an open day, or the reverse (each costs –1 point).



Harmful = a suggestion that mixes up open and closed.

Back-test = replaying 2025 using only data from before 6 Nov 2025. There: 211 harmful against 249.

No history: 36.8 against 14.2 (slide 8).

Five reasons we score higher

And one honest note on what doesn't add points.

1. **An exact rebuild of what every app shows**
   Matches Fonecta's answer key on all 22,218 days. Fonecta's sample code gets 86 of them wrong.
   22,218 of 22,218 days matched

2. **Reading two seasons back**
   Covers the years an owner didn't enter: 16% of needed changes only show up two seasons back.
   +5.9 points from this alone

3. **Learning from the true hours, not owners' entries**
   Owners enter 39% of special days but only 2% of normal days, so their entries give a skewed picture.
   39% vs 2% special vs normal days entered

4. **Recomputing similar-business statistics**
   The provided file pools 2023–2025, so in a back-test it already contains the answers. We rebuild it from data known at the time.
   No leak proved by a test

5. **Calibrated trees and a points-maximising rule**
   Gradient-boosted trees predict each day; the rule picks the action worth the most expected points on Fonecta's scoring table, with a safety margin before switching a day between open and closed.
   +7.2 points over the baseline in total

Honest note: linked holidays and habit types add no measurable points, so we don't credit them for the score.

Zero information, and still no guessing

Example: Kahvila Kipinä, a café in Turku that joined on 12 Oct 2026. No holiday history and no earlier decisions.

1. **Start from similar businesses**

   What businesses of the same type and size usually do: the habit model's starting guess for that group, plus statistics from groups of at least 10 similar businesses, computed without the future leak.

2. **Stay quiet unless it's near-certain**

   Only 1 of Kipinä's 14 special days gets a suggestion: New Year's Day closed, about 80% sure, based on similar cafés. The other 13 are left alone.

3. **Spend the one question where it counts**

   The best candidates are New Year's Day (+0.90 expected points) and Christmas Day (+0.87). New Year's Day is already on the card, so it asks about Christmas Day.

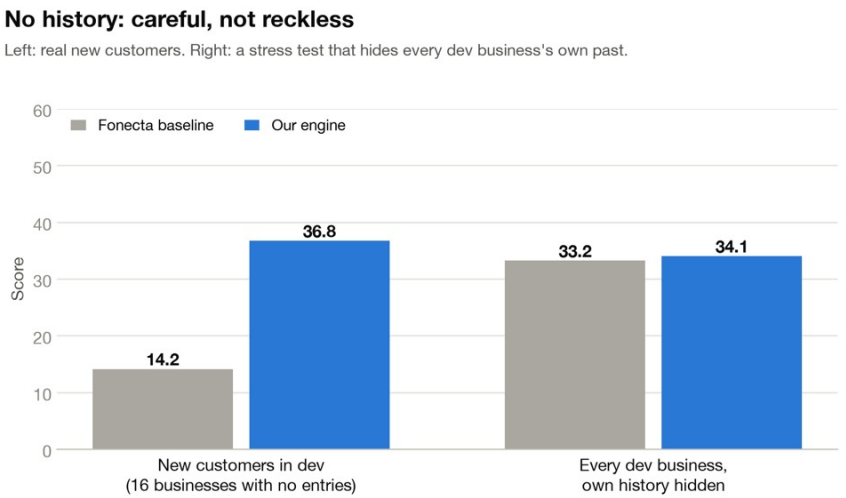
4. **Let onboarding fill the gap**

   The welcome message asks how the business handles public holidays. Two taps answer it for every public holiday, so the first card can say “as you told us” instead of guessing.

Mostly by not guessing where the baseline guesses. One answer from the owner is worth more than any amount of data from similar businesses.

36.8 vs 14.2

score on the 16 tested businesses with truly no history (Connect vs the baseline), with 2 harmful suggestions against 4



Everything happens in the owner's own chat

No app to install, no password, no form.

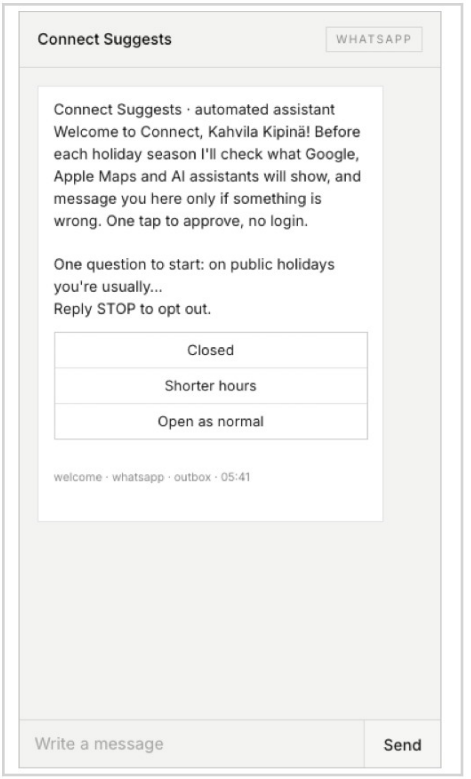
Joining takes two taps. Onboarding happens inside the owner's WhatsApp or email: no form, no password, no app.

One message per season. Connect stays silent when the listing is already right.

An answer-by date for each app, and one smart question. The question goes to the date where an answer is worth the most: +0.50 points per business, against +0.09 for a random date.

No login to decide. One tap, or a reply in the owner's own words. Claude reads it and reads it back before saving.

The demo's web dashboard only shows what happens behind the scenes. A real owner does practically everything in their chosen chat: joining, approving, editing, declining, “closed today”, questions, Undo and Stop. The only other screen is the optional no-login card page.



Onboarding in the demo: one question, three buttons.

Insert the real WhatsApp onboarding screenshot here

Placeholder: replace with a screenshot from a real phone.

Wrong suggestions are stopped in three layers

The owner always has the last word: nothing is written without their decision or their standing permission.

1 Before any message

- An exact rebuild of what each app shows today.
- A decision rule with safety margins.
- Hard rules: never contradict the owner's own entry; no past dates; valid hours only; a “no” is never suggested again; opted-out owners are never messaged.
- A business that may have closed for good goes to a person: 1 of 2 caught, zero false alarms.

2 When the owner decides

- A guessed switch between open and closed needs an explicit tick.
- A reply in the owner's own words is read back before saving.
- Links are signed and single-use. Opening a link never approves anything, so email link scanners can't approve by accident.

3 When writing to Yext

- The listing is re-read first, so the owner's own edits win.
- Only holiday hours are written; weekly hours are never touched.
- The hours are read back before the ✓ appears.
- Every step goes into a tamper-evident audit log: each record is chained to the one before it.

Every answer makes next season better

Owners' answers become remembered facts. “Every year” becomes a scoped permission, protected by the agreement check.

Owner's answer
approve, edit, decline or “every year”

→

Remembered facts
newest wins; older ones count less each season; a decline is never repeated

→

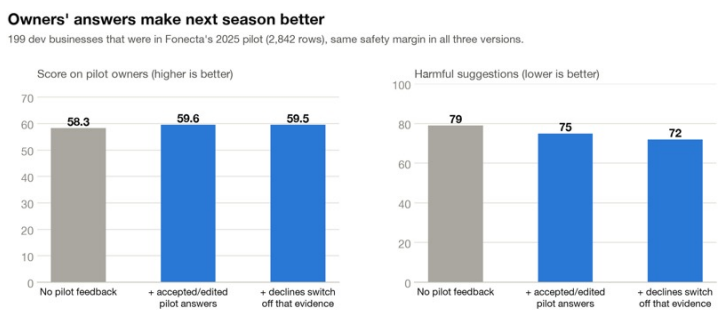
Agreement check
re-used only if this year's model gives it at least 50%

→

Next season
applied with Undo, or asked

A memory graph for every business

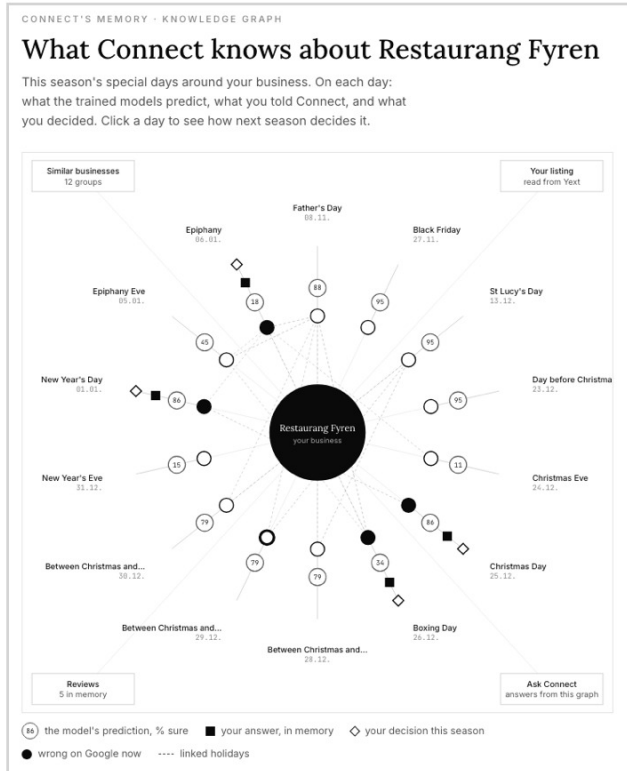
Stored in MongoDB (a database): the business's days, answers, decisions, similar businesses and reviews. Ask Connect answers owners' questions from it with graph RAG (retrieval-augmented generation over a graph): it retrieves the facts first, Claude answers only from them with a source for every sentence, and a checker rejects any date or time not in the facts.



58.3 → 59.6
score with last year's answers

79 → 72
harmful suggestions, once declines are never repeated

Measured on the owners from Fonecta's 2025 pilot.



Honest note: the prediction models don't read the graph. The graph is the memory for questions and for next season's check.

EXTENSIBILITY

One loop, many conversations

Detect → message → the owner decides → write to Yext. The same loop already runs six kinds of conversation.

Built and working

Holiday hours
the season card

Weekly hours that changed
every proposed week in testing was right

Replies to reviews
drafted for the owner to approve

“Closed today”
on Google within minutes, ends by itself

Onboarding
two taps in the owner's chat

The Autopilot note
with Undo

A new use needs only three parts



Next candidates (not built yet)

- Seasonal hours
- Event hours
- Sorting map users' suggested edits
- Phone and address changes
- Photos
- Attributes (listing details)

What it's built with, in plain words

scikit-learn gradient-boosted trees

Many small decision trees, each correcting the last, predict whether a day is closed, normal or special, and which hours.

Isotonic calibration

Adjusts the models' confidence, so “80% sure” is right about 8 times in 10.

EM habit model

Sorts businesses into 10 typical holiday habits, found with EM (expectation–maximisation), a standard method for hidden groups.

Logistic regression

A simple, explainable model that spots when a business's normal weekly hours have changed, mostly from reviews.

Value-of-information question choice

Picks the one question whose answer is worth the most expected points.

Claude Opus 5

Reads owners' replies in their own words and writes the answers; the facts always come from data.

Graph RAG over a MongoDB memory graph

Each business's facts are stored as a connected graph; answers are written only from facts retrieved from it.

Yext Knowledge API

The interface programs use to read and write listings in Yext, the service Fonecta Connect already publishes through.

WhatsApp Cloud API with a signed webhook

Meta's official way to send and receive WhatsApp messages. Replies arrive at our web address (the webhook), signed so we know they came from Meta.

Flask and HTMX

A small Python web server, and a light library that updates parts of a page without reloading it.

SHA-256 hash-chained audit log

Each record stores a fingerprint of the one before, so any later change breaks the chain and is detected.

Built to plug straight into Fonecta

Same backend, same customers, no new app.

Same Yext API as Fonecta Connect. Tested on Fonecta's own Yext test account, on the record SinceAi04, renamed Kahvila Aurora.

No new app for owners. It lives in the chat they already use: WhatsApp, email or text message.

Honest status

- Replies from a real phone work, through Meta's webhook.
- SMS (text messages) are shown on screen only.
- Review replies are drafted, not posted.
- All data is synthetic.

Going live is mostly configuration

- [ ] A production Yext key
- [ ] Fonecta's WhatsApp number and message templates
- [ ] An SMS or email provider
- [ ] Owner contacts and consent from the customer records
- [ ] One public web address
- [ ] A nightly scheduled run

Connect Suggests

Connect suggests. The owner approves once. Every year it needs them less.

61.67 vs 54.49

score on Fonecta's own scorer, against the baseline

3.5% vs 10.1%

automatic changes that get open and closed wrong, with the agreement check vs without

36.8 vs 14.2

for businesses with no history at all

Team Connect · Since AI Hackathon 2026 · Fonecta's challenge “Connect Suggests – Customer Approves”